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Inventor(s)	Weldon; David Edwin et al.

Vehicles including truck assemblies configured to selectively decouple a wheel of the truck assembly

Abstract

A vehicle including a plurality of truck assemblies is provided. Each truck assembly includes a truck axle, a first wheel fixed to the truck axle, a second wheel selectively coupled to the truck axle, a motor configured to rotate the truck axle, and a clutch operatively coupled to the second wheel. The clutch is positionable between a coupled position and a decoupled position. When the clutch is in the coupled position, the second wheel rotates with the truck axle, and when the clutch is in the decoupled position, the second wheel is free to rotate relative to the truck axle.

Inventors: **Weldon; David Edwin (Gilroy, CA), Anderson; Connor James (Alameda, CA)**

Applicant: **Volley Automation, Inc.** (Denver, CO)

Family ID: **85703490**

Assignee: **Volley Automation, Inc. (Denver, CO)**

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Primary Examiner: Neacsu; Valentin

Assistant Examiner: Stabley; Michael R

Attorney, Agent or Firm: Dinsmore & Shohl LLP

Background/Summary

TECHNICAL FIELD

(1) The present specification generally relates to vehicles for transporting a load and, more specifically, omnidirectional vehicles for including omnidirectional wheels.

BACKGROUND

(2) Omnidirectional vehicles including omnidirectional wheels such as, for example, Mecanum wheels, are capable of moving in directions other than just a forward vehicle longitudinal direction and a rearward vehicle longitudinal direction. Specifically, omnidirectional vehicles are capable of rotating about a central point of the vehicle, moving in a sideways vehicle lateral direction, and moving in a diagonal direction by independently controlling the direction of rotation of the omnidirectional wheels of the vehicle.

SUMMARY

(3) In one embodiment, a vehicle includes a plurality of truck assemblies. Each truck assembly includes: a truck axle; a first wheel fixed to the truck axle; a second wheel selectively coupled to the truck axle; a motor configured to rotate the truck axle; and a clutch operatively coupled to the second wheel, the clutch being positionable between a coupled position and a decoupled position, wherein when the clutch is in the coupled position, the second wheel rotates with the truck axle, and when the clutch is in the decoupled position, the second wheel is free to rotate relative to the truck axle.

(4) In another embodiment, a vehicle includes: a frame; a front yoke fixed to the frame; a rear yoke rotatably coupled to the frame and rotatable about a vehicle longitudinal axis; and a truck assembly rotatably coupled to opposite ends of each of the front yoke and the rear yoke, the truck assembly including: a truck axle; a first wheel fixed to the truck axle, the first wheel being an omnidirectional wheel; a second wheel selectively coupled to the truck axle; and a motor configured to rotate the truck axle.

(5) In yet another embodiment, a method includes: receiving, at a vehicle, a first instruction to perform a first maneuver, the first maneuver including moving the vehicle in a vehicle longitudinal direction, the vehicle including: a plurality of truck assemblies, each truck assembly including: a truck axle; a first wheel fixed to the truck axle; a second wheel selectively coupled to the truck axle; a motor configured to rotate the truck axle; and a clutch operatively coupled to the second wheel, the clutch being positionable between a coupled position to rotate the second wheel with the truck axle, and a decoupled position to permit the truck axle to rotate without rotating the second wheel; in response to receiving the first instruction, positioning the clutch into the coupled position; performing the first maneuver; receiving, at the vehicle, a second instruction to perform a second maneuver, the second maneuver including moving the vehicle in a non-vehicle longitudinal direction; in response to receiving the second instruction, positioning the clutch into the decoupled position; and performing the second maneuver.

(6) These and additional features provided by the embodiments described herein will be more fully understood in view of the following detailed description, in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments set forth in the drawings are illustrative and exemplary in nature and not intended to limit the subject matter defined by the claims. The following detailed description of the illustrative embodiments can be understood when read in conjunction with the following drawings, where like structure is indicated with like reference numerals and in which:

(2) FIG. **1** schematically depicts a perspective view of a vehicle, according to one or more embodiments shown and described herein;

(3) FIG. **2** schematically depicts a perspective view of a rear yoke and a pair of truck assemblies of the vehicle of FIG. **1**, according to one or more embodiments shown and described herein;

(4) FIG. **3** schematically depicts a cross-section view of a truck assembly of the pair of truck assemblies of FIG. **2**, according to one or more embodiments shown and described herein;

(5) FIG. **4** schematically depicts a perspective view of the vehicle of FIG. **1** positioned under a platform with a lifting device of the vehicle in a raised position, according to one or more embodiments shown and described herein;

(6) FIG. **5** schematically depicts an enlarged perspective view of the lifting device of FIG. **4** engaging the platform, according to one or more embodiments shown and described herein; and

(7) FIG. **6** schematically depicts a diagram of a vehicle system, according to one or more embodiments shown and described herein.

DETAILED DESCRIPTION

(8) As noted above, omnidirectional vehicles including omnidirectional wheels are capable of moving in directions other than just a forward vehicle longitudinal direction and a rearward vehicle longitudinal direction. Specifically, omnidirectional vehicles are capable of rotating about a central point of the vehicle, moving in a sideways vehicle lateral direction, and moving in a diagonal direction by independently controlling the direction of rotation of the omnidirectional wheels of the vehicle. However, the amount of load the vehicle is configured to carry is limited by the number of omnidirectional wheels of the vehicle. Providing additional wheels increases the load carrying capacity of the vehicle, but increases the amount of friction on the wheels during movement in a non-vehicle longitudinal direction, thereby reducing the life span of the wheels. One solution to this problem is to independently adjust the rotational speed of each wheel to account for the position of each wheel on the vehicle. However, this requires each wheel to have a separate motor, which increases the weight of the vehicle and increases the potential for part failure.

(9) Embodiments provided herein are directed to improved omnidirectional vehicles that reduce friction on the wheels of the vehicle while moving in a non-vehicle longitudinal direction without increasing the weight of the vehicle and the potential for part failure by including a designated motor for each wheel.

(10) Embodiments described herein are directed to an omnidirectional vehicle including a plurality of truck assemblies and each truck assembly configured to selectively decouple one of a pair of wheels of the truck assembly when moving the vehicle in a non-vehicle longitudinal direction. By decoupling one of the wheels of each truck assembly, the decoupled wheel is able to freely rotate. Various embodiments of the vehicle and the operation of the vehicle are described in more detail herein. Whenever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts.

(11) Referring now to FIG. **1**, a vehicle **100** is illustrated according to one or more embodiments described herein. As described in more detail herein, the vehicle **100** is an omnidirectional land-based vehicle capable of moving in any direction by controlling the direction of rotation of individual omnidirectional wheels of the vehicle **100**. As described herein, the vehicle **100** may be suitable for carrying large loads and transporting the payload across a vehicle transit surface.

However, it should be appreciated that the vehicle **100** disclosed herein may be suitable for any other purpose without deviating from the scope of the present disclosure.

(12) Referring now to FIG. **1**, the vehicle **100** may generally include a frame **102**, a front yoke **104**, and opposite rear yoke **106** provided at opposite ends of the frame **102**, and a truck assembly **108** mounted at opposite ends of each of the front yoke **104** and the rear yoke **106**. Each truck assembly **108** includes a first or outer wheel **110**, a second or inner wheel **112**, and a motor **114** selectively coupled to at least one of the outer wheel **110** and the inner wheel **112**. As used herein, the term "truck assembly" refers to an assembly generally including a pair of wheels operatively coupled to a motor to transport a vehicle. As discussed herein, a vehicle may include a plurality of truck assemblies, such as four truck assemblies, arranged at various corners, sides, or other suitable locations of the vehicle.

(13) The frame **102** includes a front rail **116**, an opposite rear rail **118**, a first side rail **120**, and an opposite second side rail **122** extending between the front rail **116** and the rear rail **118**. The frame **102** defines an open interior **124** between the front rail **116**, the rear rail **118**, the first side rail **120**, and the second side rail **122** in which various components of the vehicle **100** may be stowed. For example, and as discussed in more detail herein, the vehicle **100** may include one or more batteries, control systems, motors, sensors, and the like provided within the open interior **124** of the frame **102**. The frame **102** of the vehicle **100** may further include a plurality of reinforcing beams **126** extending parallel to either a vehicle longitudinal axis L**1** between the front rail **116** and the rear rail **118**, or a vehicle lateral axis L**2** between the first side rail **120** and the second side rail **122**. The vehicle longitudinal axis L**1** extends parallel to the +/−Y axis of the coordinate axes of the coordinate axes depicted in FIG. **1**. The vehicle lateral axis L**2** is transverse to the vehicle longitudinal axis L**1** and extends parallel to the +/−X axis of the coordinate axes depicted in FIG. **1**. The reinforcing beams **126** may be suitable for reinforcing the frame **102** and providing a surface to which the various components provided within the open interior **124** of the frame **102** may be mounted.

(14) As described in more detail herein, the vehicle **100** includes a lifting assembly **128** for engaging an object to be transported by the vehicle **100**. In embodiments, the lifting assembly **128** includes a plurality of lifting devices **130** pivotally fixed to an upper surface **132** of the frame **102**, particularly the first side rail **120** and the second side rail **122**. As shown in FIG. **1**, the lifting assembly **128** includes a pair of lifting devices **130** pivotally fixed to the first side rail **120** at respective proximal ends thereof, and a pair of lifting devices **130** pivotally fixed to the second side rail **122** at respective proximal ends **134** thereof. However, it should be appreciated that the lifting assembly **128** may include any number of lifting devices **130** such as, for example, one lifting device **130** or more than five lifting devices **130**. Additionally, although each of the lifting devices **130** are illustrated as being pivotally fixed to the first side rail **120** or the second side rail **122**, it should be appreciated that the lifting devices **130** may alternatively be pivotally fixed to either the front rail **116** or the rear rail **118**. Moreover, the lifting assembly **128** may include any other suitable lifting device movable in a non-pivoting motion, such as a vertically translating or elevating device mounted at any suitable location of the vehicle **100** such as, for example, within the open interior **124** of the frame **102**.

(15) In the present embodiment, each lifting device **130** is pivotally coupled to the first side rail **120** or the second side rail **122** by a lifting pivot **136** extending through the proximal end **134** of the lifting device **130**. The lifting device **130** may be operated by a control system to pivot about the lifting pivot **136** between a lowered position (FIG. **1**) and a raised position (FIG. **4**). When in the lowered position, the lifting device **130** is rotated in the direction of arrow A**1** about the lifting pivot **136** such that the lifting device **130** is lowered to extend substantially parallel to the first side rail **120** and the second side rail **122**. When in the raised position, the lifting device **130** is rotated in the direction of arrow A**2** about the lifting pivot **136** such that the lifting device **130** is raised to contact an object to be transported such as, for example, a platform carrying a load. As described in

more detail herein, the lifting device **130** may include a contact member **138** provided at a distal end **140** opposite the proximal end **134** to contact and engage the object. In embodiments, each of the lifting devices **130** may be operated such that the contact member **138** of each lifting device **130** simultaneously contacts and engages the object to be transported.

(16) The front yoke **104** is fixedly mounted to the front rail **116** of the frame **102** of the vehicle **100** such that the front yoke **104** does not rotate in any direction. To the contrary, the rear yoke **106** is rotatably coupled to the rear rail **118** of the frame **102** of the vehicle **100**. In embodiments, the frame **102** of the vehicle **100** includes a rear housing **142** fixed to the rear rail **118** and extending in a direction opposite the front rail **116**. The rear housing **142** defines a housing cavity **144** through which the rear yoke **106** extends and the rear yoke **106** is rotatably fixed to the rear housing **142** by a rear yoke pivot **146** extending coaxial with the vehicle longitudinal axis L**1**. Accordingly, the rear yoke **106** is configured to rotate about the vehicle longitudinal axis L**1** at the rear yoke pivot **146** in the direction of arrow B**1** and arrow B**2**. As described in more detail herein, the truck assemblies **108** are rotatably coupled to the front yoke **104** and the rear yoke **106** to allow for the outer wheel **110** and the inner wheel **112** of each truck assembly **108** to move in opposite vehicle vertical directions. As such, each truck assembly **108** rotates in the direction of arrow C**1** and arrow C**2** about a respective truck assembly axis L**3** extending parallel to the vehicle longitudinal axis L**1** and the +/−Y axis of the coordinate axes depicted in FIG. **1**. Rotation of the rear yoke **106** about the vehicle longitudinal axis L**1** and each truck assembly **108** about the respective truck assembly axis L**3** allows the outer wheel **110** and the inner wheel **112** to maintain continuous contact with the vehicle transit surface while in use.

(17) The vehicle **100** includes a total of eight wheels, four outer wheels **110** and four inner wheels **112**. As referred to herein, the outer wheel **110** is spaced apart from the vehicle longitudinal axis L**1** by a first distance and the inner wheel **112** is spaced apart from the vehicle longitudinal axis L**1** by a second distance less than the first distance. Stated another way, the inner wheels **112** are located closer to the vehicle longitudinal axis L**1** than the outer wheels **110**. As shown in FIG. **1**, the outer wheel **110** and the inner wheel **112** of each truck assembly **108** is illustrated as an omnidirectional wheel and, specifically, a Mecanum wheel. However, it should be understood that, in embodiments, only one of the outer wheel **110** and the inner wheel **112** may be an omnidirectional wheel and, more specifically, a Mecanum wheel. As used herein, the term “omnidirectional wheel” refers to wheels with small discs or rollers provided around a circumference which is non-parallel to a turning direction of the wheel. This allows the wheel to be driven with full force, but also slide at an angle relative to the driving direction. As referred to herein, the outer wheels **110** will be referred to as the omnidirectional wheels, such as Mecanum wheels, while the inner wheels **112** may be either omnidirectional wheels, such as Mecanum wheels, or any other suitable unidirectional wheel or rolling device. In embodiments, in which only the outer wheel **110** is an omnidirectional wheel, the outer wheel **110** of each truck assembly **108** is an omnidirectional wheel. Alternatively, in embodiments in which only the inner wheel **112** is an omnidirectional wheel, the inner wheel **112** of each truck assembly **108** is an omnidirectional wheel. Providing both an outer wheel **110** and an inner wheel **112** for each truck assembly **108** increases the load carrying capacity of the vehicle **100** as compared to vehicles including only four wheels. As noted above, utilizing four omnidirectional wheels, for example, Mecanum wheels, as the outer wheels **110** allows the vehicle **100** to move in any direction by independently controlling the direction of rotation of the outer wheels **110**. However, rotating each of the inner wheels **112** along with the outer wheels **110** results in increased friction against the vehicle transit surface and increased wear on the wheels **110**, **112**, thereby reducing the overall life of the wheels **110**, **112**. Accordingly, the vehicle **100** is configured to decouple the inner wheels **112** of each truck assembly **108** when moving in any direction other than a vehicle forward direction or a vehicle rearward direction parallel to the vehicle longitudinal axis L**1**, such as when the vehicle **100** is performing a rotation maneuver or a sideways maneuver, thereby permitting the outer wheels **110** to execute the

omnidirectional movements.

(18) Operation of the truck assemblies **108** may result in pushing and pulling forces being applied at opposite ends of the front yoke **104** and the rear yoke **106** relative to the frame **102** of the vehicle **100**. As such, to prevent damage to the vehicle **100** and unintended movement at the front yoke **104** and the rear yoke **106** relative to the frame **102**, the vehicle **100** may include a linkage assembly **148** coupling the front yoke **104** and the rear yoke **106** to the frame **102**. In embodiments, as shown in FIGS. **1** and **4**, the linkage assembly **148** may include a plurality of tie rods **150** coupling the rear yoke **106** to the frame **102**, specifically the first side rail **120** and the second side rail **122**. Although only a first pair of tie rods **150** are shown in FIG. **1** extending between an end of the rear yoke **106** and the first side rail **120**, it should be appreciated that a second pair of tie rods **150** may similarly be provided to extend between an opposite end of the rear yoke **106** and the second side rail **122**. As such, the pair of tie rods **150** extending along the first side rail **120** and the pair of tie rods **150** extending along the second side rail **122** coupling the frame **102** to the rear yoke **106** forms a linkage assembly **148**. In embodiments, a similar linkage assembly **148** may be provided to couple the first side rail **120** and the second side rail **122** to the front yoke **104** as well.

(19) Referring now to FIG. **2**, the rear yoke **106** is illustrated. It should be appreciated that the rear yoke **106** and the front yoke **104** are identical in structure. Because the rear yoke **106** and the front yoke **104** are identical in structure, a plurality of mounting holes **152** may be provided on a frame facing surface **154** of the rear yoke **106** and configured to receive respective mounting shafts. Accordingly, the mounting shafts may be provided in the mounting holes **152** of the front yoke **104** to fixedly mount the front yoke **104** to the front rail **116** of the frame **102**. For the rear yoke **106**, the mounting holes **152** may not be utilized. Additionally, the rear yoke **106** includes a central aperture **156** formed in the frame facing surface **154** of the rear yoke **106**. The central aperture **156** is configured to receive the rear yoke pivot **146** (FIG. **1**) and rotatably couple the rear yoke **106** to the rear housing **142** (FIG. **1**). Because the rear yoke **106** and the front yoke **104** are identical in structure, the front yoke **104** may also include a central aperture formed therein, but may not utilized as the front yoke **104** is not rotatably coupled to the front rail **116**. The rear yoke pivot **146** rotatably couples the rear yoke **106** to the rear housing **142** such that the rear yoke **106** is permitted to rotate about the vehicle longitudinal axis L**1** in the direction of arrow B**1** and arrow B**2**. By permitting the rear yoke **106** to rotate relative to the rear rail **118** of the frame **102** of the vehicle **100**, the relative height positions of the truck assemblies **108** provided at opposite ends of the rear yoke **106** are adjusted so that the outer wheel **110** and the inner wheel **112** of each truck assembly **108** are able to maintain continuous contact with the vehicle transit surface.

(20) Referring still to FIG. **2**, a cross-section view of one of the truck assemblies **108** mounted onto the rear yoke **106** is shown. In embodiments, the truck assembly **108** has a yoke receptacle **158** defining a receptacle cavity **160** through which the rear yoke **106** extends. The truck assembly **108** includes a truck assembly shaft **162** rotatably coupling the truck assembly **108** to the rear yoke **106**. In embodiments, the truck assembly shaft **162** may be received within and fixed to a truck bore **164** formed in the truck assembly **108**. Additionally, the truck assembly shaft **162** extends through a portion of the rear yoke **106** that is received within the yoke receptacle **158** of the truck assembly **108**. Thus, as discussed herein, the truck assembly shaft **162** permits rotation of the truck assembly **108** in the direction of arrow C**1** and arrow C**2** about the truck assembly axis L**3**.

(21) The rear yoke **106** may include a first pair of dampers **166** extending between the rear yoke **106** and a respective truck assembly **108**. The dampers **166** are configured to provide a biasing force to cause the truck assemblies **108** to return from a rotated position to an initial position. For example, when the truck assembly **108** rotates about the truck assembly shaft **162** in the direction of arrow C**1**, the respective damper **166** will apply a biasing force in the direction of arrow D**1** to return the truck assembly **108** to the initial position. Alternatively, when the truck assembly **108** rotates about the truck assembly shaft **162** in the direction of arrow C**2**, the respective damper **166** will apply a biasing force in the direction of arrow D**2** to return the truck assembly **108** to the initial

position.

(22) Although not shown, it should be appreciated that the truck assemblies **108** at the front yoke **104** are mounted to the front yoke **104** in the same manner utilizing a similar truck assembly shaft. Similarly, as shown in FIG. **1**, a second pair of dampers **166** is also provided at the front yoke **104** for assisting the truck assemblies **108** to return to an initial position.

(23) Referring now to FIG. **3**, a cross-section view of a truck assembly **108** is illustrated. Each truck assembly **108** is identical in structure. As discussed herein, the truck assembly **108** includes the outer wheel **110** and the inner wheel **112**. The outer wheel **110** includes an outer wheel bore **168**, and the inner wheel **112** includes an inner wheel bore **170**. The truck assembly **108** further includes the motor **114**, a gearbox **174**, and a truck axle **176** extending through a gearbox bore **178** formed in the gearbox **174**. The motor **114** is communicatively coupled to the gearbox **174** to rotate the truck axle **176** in the directions of either arrow E**1** or arrow E**2** about a truck axle axis L**4** extending transverse to the vehicle longitudinal axis (FIG. **1**). In embodiments, the outer wheel **110** is fixed to the truck axle **176** such that operation of the motor **114** in a first state rotates the truck axle **176** in the direction of arrow E**1** and similar rotates the outer wheel **110** in the same direction. Alternatively, the motor **114** may be operated in a second state to rotate the truck axle **176** and the outer wheel **110** in the direction of arrow E**2**.

(24) In embodiments, the inner wheel **112** is selectively fixed to the truck axle **176** such that the inner wheel **112** may be configured to rotate with the outer wheel **110** in the same direction. More particularly, the truck assembly **108** includes a clutch **180** operatively coupled to the inner wheel **112** and operable between a coupled position and a decoupled position and provided within the inner wheel **112**. When the clutch **180** is in the coupled position, the inner wheel **112** engages or becomes rotatably fixed to the truck axle **176** such that rotation of the truck axle **176** causes the inner wheel **112** to rotate in the same direction. Alternatively, when the clutch **180** is in the decoupled position, the inner wheel **112** disengages the truck axle **176** such that the truck axle **176** may rotate within the inner wheel bore **170** of the inner wheel **112** without causing the inner wheel **112** to rotate. It should be appreciated that any controllable clutch may be suitable for coupling and decoupling the inner wheel **112** to the truck axle **176**.

(25) In embodiments, the clutch **180** is an electromagnetic clutch including a terminal **182**. In embodiments, when a voltage is applied to the terminal **182**, the clutch **180** is positioned into either the coupled position or the decoupled position. In embodiments, the clutch **180** may include a pair of toothed members **184** for coupling the inner wheel **112** to the truck axle **176**. Upon being actuated, the toothed members **184** may be configured to translate along the truck axle axis L**4** to engage or disengage various parts of the truck axle **176** and/or the inner wheel **112**. For example, the toothed members **184** may be configured to slide along the truck axle axis L**4** in the direction of either arrow F**1** or arrow F**2** to engage or disengage various parts of the truck axle **176** and/or the inner wheel **112**. In some embodiments, the toothed members **184** may be configured to move in the same direction such as in the direction of arrow F**1** or arrow F**2**. In other embodiments, the toothed members **184** may be configured to move in opposite directions relative to one another when alternating between the coupled position and the decoupled position.

(26) Although reference is made herein to the outer wheel **110** being fixed to the truck axle **176** and the inner wheel **112** being configured to engage and disengage the truck axle **176**, it should be appreciated that, in embodiments, the inner wheel **111** is configured to engage and disengage the truck axle **176** while the outer wheel **110** remains fixed to the truck axle **176**. In such an embodiment, the clutch **180** is positioned within the inner wheel **112** and configured to operate between the coupled position and the decoupled position to engage and disengage the truck axle **176** such that the inner wheel **112** is configured to freely rotate relative to the truck axle **176** when the clutch **180** is in the decoupled position. In other embodiments, the clutch **180** may be positioned outside of the inner wheel **112**.

(27) Referring now to FIG. **4**, the vehicle **100** is illustrated with one of the lifting devices **130** of the

lifting assembly **128** in the raised position and engaging an object such as, for example, a platform **186**. As shown in FIG. **4**, the vehicle **100** is positioned under the platform **186**, such as by operating the truck assemblies **108** to maneuver the vehicle **100** to be located under the platform **186**. In embodiments, the platform **186** may be utilized to support objects such as, for example, additional vehicles, equipment, storage units, and the like. Once the vehicle **100** is determined to be in position under the platform **186**, the lifting assembly **128** is operated to cause each of the lifting devices **130** to move from the lowered position into the raised position and engage the platform **186**. In embodiments, each lifting device **130** of the lifting assembly **128** may include a roller **188** extending between the distal end **140** of the lifting device **130** and a respective one of the first side rail **120** and the second side rail **122**. As the roller **188** is operated to extend from the first side rail **120** or the second side rail **122**, the lifting device **130** is configured to pivot at the lifting pivot **136** extending through the proximal end **134** of the lifting device **130**, as shown in FIG. **1**. Pivoting of the lifting device **130** causes the lifting device **130** to be positioned within the raised position and contact the platform **186**.

(28) More specifically, the each lifting device **130** includes the contact member **138** provided at the distal end **140** of the lifting device **130**. In embodiments, the contact member **138** has a convex outer surface **192** extending in a direction opposite the respective one of the first side rail **120** and the second side rail **122**. The contact member **138** may be configured to engage a receiving member of the platform **186**. For example, as shown in FIG. **5**, the platform **186** includes a receiving member **194** fixed to a bottom surface **196** of the platform **186**. The receiving member **194** has a planar surface **198** facing the bottom surface **196** of the platform **186** and a concave outer surface **195** opposite the planar surface **198**. As such, the receiving member **194** is shaped to receive the contact member **138** of the lifting device **130**, specifically the convex outer surface **192**. The contact member **138** may be integrally formed on an upper surface **199** of the distal end **140** of the lifting device **130**. Accordingly, as shown in FIG. **5**, the roller **188** may positioned to be received within the distal end **140** of the lifting device **130** underneath the contact member **138** so as to apply a force on the distal end **140** of the lifting device **130**.

(29) As the force is applied to the distal end **140** of the lifting device **130** by the roller **188**, the lifting device **130** moves toward the raised position so that the contact member **138** of the lifting device **130** is received within the receiving member **194** of the platform **186**. It should be appreciated that the platform **186** includes a plurality of receiving members **194** with each receiving member **194** located to engage a respective contact member **138** when the lifting devices **130** are each in the raised position. As such, the platform **186** may be supported at four locations when positioned over the vehicle **100**. The platform **186** may then be raised by the vehicle **100** and transported along with the vehicle **100** to a target destination. In embodiments, it should be appreciated that the contact member **138** may alternatively include a concave outer surface and the receiving member **194** may have a convex outer surface so as to receive the contact member **138**. In addition, alternative engagements between the contact member **138** and the receiving member **194** are within the scope of the present disclosure such as, for example, engagement by way of locking mechanisms, magnetic interfaces, and the like.

(30) Referring now to FIG. **6**, a vehicle system **200** may be provided for operating the various components of the vehicle **100**, for example, the truck assemblies **108** to couple and decouple the inner wheel **112** and position the vehicle **100**, and the lifting assembly **128** to position the lifting devices **130** between the raised position and the lowered position. The vehicle system **200** may comprise a controller **202**, a power supply **204**, a sensing device **206**, the truck assemblies **108**, the lifting assembly **128**, network interface hardware **208**, and a communication path **210** communicatively coupled these components.

(31) The controller **202** may comprise a processor **212** and a non-transitory electronic memory **214** to which various components are communicatively coupled. In some embodiments, the processor **212** and the non-transitory electronic memory **214** and/or the other components are included within

a single device. In other embodiments, the processor **212** and the non-transitory electronic memory **214** and/or the other components may be distributed among multiple devices that are communicatively coupled. The controller **202** may include non-transitory electronic memory **214** that stores a set of machine-readable instructions. The processor **212** may execute the machine-readable instructions stored in the non-transitory electronic memory **214**. The non-transitory electronic memory **214** may comprise RAM, ROM, flash memories, hard drives, or any device capable of storing machine-readable instructions such that the machine-readable instructions can be accessed by the processor **212**. Accordingly, the vehicle system **200** described herein may be implemented in any computer programming language, as pre-programmed hardware elements, or as a combination of hardware and software components. The non-transitory electronic memory **214** may be implemented as one memory module or a plurality of memory modules.
(32) In some embodiments, the non-transitory electronic memory **214** includes instructions for executing the functions of the vehicle system **200**. The instructions may include instructions for operating the motor **114** of one or more of the truck assemblies **108** in either the first state or the second state, positioning the clutch **180** of the truck assemblies **108** between the coupled state and the decoupled state to couple or decouple the inner wheel **112** of each of the truck assemblies **108** to the truck axle **176**, positioning the lifting devices **130** between the raised state and the lowered state, and the like. More particularly, the instructions may include automatically positioning the clutch **180** of the truck assemblies **108** into the decoupled state when the vehicle **100** is moving or instructed to move in any direction other that in the vehicle longitudinal direction.
(33) The processor **212** may be any device capable of executing machine-readable instructions. For example, the processor **212** may be an integrated circuit, a microchip, a computer, or any other computing device. The non-transitory electronic memory **214** and the processor **212** are coupled to the communication path **210** that provides signal interconnectivity between various components and/or modules of the vehicle system **200**. Accordingly, the communication path **210** may communicatively couple any number of processors with one another, and allow the modules coupled to the communication path **210** to operate in a distributed computing environment. Specifically, each of the modules may operate as a node that may send and/or receive data. As used herein, the term “communicatively coupled” means that coupled components are capable of exchanging data signals with one another such as, for example, electrical signals via conductive medium, electromagnetic signals via air, optical signals via optical waveguides, and the like.
(34) As schematically depicted in FIG. **6**, the communication path **210** communicatively couples the processor **212** and the non-transitory electronic memory **214** of the controller **202** with a plurality of other components of the vehicle system **200**. For example, the vehicle system **200** depicted in FIG. **6** includes the processor **212** and the non-transitory electronic memory **214** communicatively coupled with the power supply **204**, the sensing device **206**, the truck assemblies **108**, and the lifting assembly **128**.
(35) The power supply **204** (e.g., battery) provides power to the various components of the vehicle **100** such as, for example, the sensing device **206**, the truck assemblies **108**, and the lifting assembly **128**. In some embodiments, the power supply **204** is a rechargeable direct current power supply. It is to be understood that the power supply **204** may be a single power supply or battery for providing power to the sensing device **206**, the truck assemblies **108**, and the lifting assembly **128**.
(36) The sensing device **206** is configured to detect a surrounding environment of the vehicle **100** to identify a location of the vehicle **100**. The sensing device **206** may include one or more imaging sensors configured to operate in the visual and/or infrared spectrum to sense visual and/or infrared light. Additionally, while the particular embodiments described herein are described with respect to hardware for sensing light in the visual and/or infrared spectrum, it is to be understood that other types of sensors are contemplated. For example, the systems described herein could include one or more Light Detection and Ranging (LIDAR) sensors, radar sensors, sonar sensors, or other types of sensors and that such data could be integrated into or supplement the data collection described

herein. Ranging sensors like radar may be used to obtain a rough depth and speed information of an object. In embodiments, the vehicle **100** may include a plurality of sensing devices **206** arranged at various locations of the vehicle **100** such as, for example, at the front rail **116**, the rear rail **118**, the first side rail **120**, and the second side rail **122**. The vehicle **100** may also include one or more sensing devices **206** located at a bottom surface of the vehicle **100** so as to detect a surface over which the vehicle **100** moves along. The sensing device **206** may be configured to identify when the vehicle **100** is in position relative to an object, such as the platform **186**, and, when in the correct position, activate the lifting assembly **128** to contact and engage the platform **186**.
(37) In some embodiments, the vehicle system **200** includes network interface hardware **208** for communicatively coupling the vehicle system **200** to a portable device **216** via a network **218**. The portable device **216** may include, without limitation, a smartphone, a tablet, a personal media player, or any other electric device that includes wireless communication functionality. The portable device **216** is capable of communicating with the network interface hardware **208**, utilizing Wi-Fi, Bluetooth, and/or any other suitable communication protocol. It is to be appreciated that, when provided, the portable device **216** may serve to wirelessly provide user commands to the controller **202**. As such, a user may be able to control or set a program for controlling the vehicle **100**. Thus, the vehicle **100** may be controlled remotely via the portable device **216** wirelessly communicating with the controller **202** via the network **218**.
(38) From the above, it is to be appreciated that defined herein is an omnidirectional vehicle including a plurality of truck assemblies. Each truck assembly includes an outer wheel, an inner wheel, and a clutch configured to selectively decouple one of the outer wheel and the inner wheel when moving the vehicle in a non-vehicle longitudinal direction. By decoupling one of the wheels of each truck assembly, the decoupled wheel is able to freely rotate relative to the other wheel.
(39) While particular embodiments have been illustrated and described herein, it should be understood that various other changes and modifications may be made without departing from the scope of the claimed subject matter. Moreover, although various aspects of the claimed subject matter have been described herein, such aspects need not be utilized in combination. It is therefore intended that the appended claims cover all such changes and modifications that are within the scope of the claimed subject matter.

Claims

1. A vehicle comprising: a plurality of truck assemblies, each truck assembly comprising: a truck axle; a first wheel fixed to the truck axle; a second wheel selectively coupled to the truck axle; a motor configured to rotate the truck axle; and a clutch operatively coupled to second wheel, the clutch being positionable between a coupled position and a decoupled position; a frame comprising a front rail, a rear rail, and a first side rail extending between the front rail and the rear rail; a rear yoke rotatably coupled to the rear rail and rotatable relative to the rear rail about a vehicle longitudinal axis; and a linkage assembly extending between the rear yoke and the plurality of truck assemblies mounted to the rear yoke, the linkage assembly comprising: a first pair of tie rods extending between the first side rail and a first end of the rear yoke; and a second pair of tie rods extending between the first side rail and a second end of the rear yoke, wherein when the clutch is in the coupled position, the second wheel rotates with the truck axle, and when the clutch is in the decoupled position, the second wheel is free to rotate relative to the truck axle.
2. The vehicle of claim 1, wherein: the frame further comprises: a second side rail extending between the front rail and the rear rail, and a front yoke fixed to the front rail.
3. The vehicle of claim 2, wherein the first wheel is spaced apart from the vehicle longitudinal axis by a first distance and the second wheel is spaced apart from the vehicle longitudinal axis by a second distance less than the first distance.
4. The vehicle of claim 3, wherein the second wheel of each truck assembly of the plurality of truck

assemblies is an omnidirectional wheel.

5. The vehicle of claim 2, wherein each truck assembly of the plurality of truck assemblies are rotatably coupled at opposite ends of each of the front yoke and the rear yoke to rotate between an initial position and a rotated position.

6. The vehicle of claim 5, wherein the front yoke includes a front pair of dampers extending between the front yoke and a respective truck assembly, and the rear yoke includes a rear pair of dampers extending between the rear yoke and a respective truck assembly, the front pair of dampers and the rear pair of dampers providing a biasing force to return the truck assemblies to the initial position from the rotated position.

7. The vehicle of claim 2, further comprising a lifting assembly including a plurality of lifting devices positionable between a lowered position and a raised position, wherein when in the lowered position, each of the plurality of lifting devices are positioned along an upper surface of one of the first side rail and the second side rail, and wherein when in the raised position, each of the plurality of lifting devices are pivoted at a proximal end of the plurality of lifting devices.

8. The vehicle of claim 7, wherein each lifting device of the plurality of lifting devices includes a contact member provided at a distal end opposite the proximal end for engaging a receiving member of a platform when the vehicle is positioned under the platform.

9. The vehicle of claim 1, wherein the first wheel of each truck assembly of the plurality of truck assemblies is an omnidirectional wheel.

10. The vehicle of claim 9, wherein the second wheel of each truck assembly of the plurality of truck assemblies is an omnidirectional wheel.

11. A method comprising: receiving, at a vehicle, a first instruction to perform a first maneuver, the first maneuver including moving the vehicle in a vehicle longitudinal direction, the vehicle comprising: a plurality of truck assemblies, each truck assembly comprising: a truck axle; a first wheel fixed to the truck axle; a second wheel selectively coupled to the truck axle; a motor configured to rotate the truck axle; and a clutch operatively coupled to the second wheel, the clutch being positionable between a coupled position to rotate the second wheel with the truck axle, and a decoupled position to permit the truck axle to rotate without rotating the second wheel; a frame comprising a front rail, a rear rail, and a first side rail extending between the front rail and the rear rail; a rear yoke rotatably coupled to the rear rail and rotatable relative to the rear rail about a vehicle longitudinal axis; and a linkage assembly extending between the rear yoke and the plurality of truck assemblies mounted to the rear yoke, the linkage assembly comprising: a first pair of tie rods extending between the first side rail and a first end of the rear yoke; and a second pair of tie rods extending between the first side rail and a second end of the rear yoke, in response to receiving the first instruction, positioning the clutch into the coupled position; performing the first maneuver; receiving, at the vehicle, a second instruction to perform a second maneuver, the second maneuver including moving the vehicle in a non-vehicle longitudinal direction; in response to receiving the second instruction, positioning the clutch into the decoupled position; and performing the second maneuver.

12. The method of claim 11, wherein: the first wheel is spaced apart from a vehicle longitudinal axis by a first distance and the second wheel is spaced apart from the vehicle longitudinal axis by a second distance less than the first distance; and the first wheel and the second wheel of each truck assembly of the plurality of truck assemblies are omnidirectional wheels.

13. A vehicle comprising: a plurality of truck assemblies, each truck assembly comprising: a truck axle; a first wheel fixed to the truck axle; a second wheel selectively coupled to the truck axle; a motor configured to rotate the truck axle; and a clutch operatively coupled to second wheel, the clutch being positionable between a coupled position and a decoupled position; a lifting assembly including a plurality of lifting devices positionable between a lowered position and a raised position, wherein when in the raised position, each of the plurality of lifting devices are pivoted at a proximal end of the plurality of lifting devices, wherein each lifting device of the plurality of lifting

devices includes a contact member provided at a distal end opposite the proximal end for engaging a receiving member of a platform when the vehicle is positioned under the platform, wherein when the clutch is in the coupled position, the second wheel rotates with the truck axle, and when the clutch is in the decoupled position, the second wheel is free to rotate relative to the truck axle.